Existence and regularity of the minima of integral shape functionals

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Shape Optimization: Introduction

A shape optimization problem is a variational problem of the form

$$\min\{\mathcal{F}(\Omega) \ : \ \Omega \in \mathcal{A}\}.$$

The functional $\mathcal{F}(\Omega)$ depends on the solution of a PDE in Ω, e.g.

$$\mathcal{F}(\Omega) = \frac{\int_\Omega |\nabla u|^2 \, dx}{\int_\Omega |u|^2 \, dx},$$

where u is a solution to the eigenvalue problem

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Additional terms such as the perimeter or measure of Ω are often involved.

Shape Optimization: Introduction

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The family of admissible sets $\mathcal{A}$ can be:

- **open** (or quasi-open) sets contained in a box $D \subseteq \mathbb{R}^d$;
- **rectifiable** sets with $\mathcal{H}^{d-1}(\partial\Omega)$ finite contained in $D \subseteq \mathbb{R}^d$;
- **convex** sets contained in $D \subseteq \mathbb{R}^d$;
- sets with **Lipschitz boundary** (or C^1) contained in $D \subseteq \mathbb{R}^d$.

Additional constraints on $|\Omega|$ or $\mathrm{Per}(\Omega)$ are sometimes imposed.

Shape Optimization: Main Examples

- **Spectral** functionals such as

$$\mathcal{F}(\Omega) = F(\lambda_1(\Omega), \ldots, \lambda_k(\Omega)).$$

For example, if $\lambda_i(\Omega)$ are Dirichlet eigenfunctions and $\Omega \subseteq D$ with $|\Omega| \leq 1$, then there is a robust existence theory [BD93] when D is bounded.

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- **Integral** functionals such as

$$\mathcal{F}(\Omega) = \int_\Omega j(x, u_\Omega)\, dx,$$

where u_Ω is the solution of a given PDE on Ω. This is a typical situation in Optimal Control or models in Physics/Engineering.

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Shape Optimization: Problems in Control Form

Let $D \subset \mathbb{R}^d$ be an open set of finite measure. For $A \subset D$, the PDE providing the state function u_A of the system under observation is

$$-\Delta_p u_A = f \text{ in } A, \qquad u_A = 0 \text{ in } \bar{D} \setminus A. \tag{1}$$

In [BMV21], we focus on optimization problems of the form

$$\min \{\mathcal{F}(A) \ : \ A \text{ open}, \ A \subset D\}, \tag{2}$$

where $\mathcal{F}$ is, for example, the affine model functional:

$$\mathcal{F}(A) = -\int_D g(x) u_A \, dx + |A|. \tag{3}$$

Main Result

Theorem (Buttazzo, M., Velichkov)

Let $f \geq 0$ be a function in $L^q(D)$, for some $q > d/p$ and $q \geq 1$. Let $g \geq 0$ be measurable function in $L^r(D)$, for some $r > 1$, and let

$$\mathcal{F}(A) = -\int_D g(x) u_A \, dx + |A|.$$

Then the following holds:

(a) If there is $C > 0$ such that

$$f(x) \leq C g(x) \qquad \text{for every } x \in D,$$

then there is an open set $\Omega = \{\bar{u} > 0\} \subset D$ solution to (2).

(b) Every solution to (2) has finite perimeter.

Remarks

- When f and g are proportional, the problem reduces to a free boundary problem and the optimal state function $\bar{u}$ solves

$$\min_{u \in W_0^{1,p}(D)} \left\{ \frac{1}{p} \int_D |\nabla u|^p \, dx - \int_D f(x) u \, dx + \lambda \frac{p-1}{p} \, |\{u > 0\}| \right\}.$$

  The regularity of the free boundary $\partial\{\bar{u} > 0\}$ is obtained in [DP05] when $f = 0$; only partial results are available [BS20] when $f \geq 0$.

- When f and g are not proportional, given Ω optimal shape, the corresponding state function u_Ω is a priori optimal only among functions that serve as state functions on other domains.

Idea of the Proof for $p > d$

The same proof works in a larger class of admissible sets, e.g.

$$\mathcal{A} = \{\Omega \subset D \ : \ \Omega \text{ open and } |\Omega| \leq m\}\,.$$

Let Ω_n be a minimizing sequence and u_n the corresponding state functions. By applying the Poincaré inequality and Sobolev embedding, we establish that:

$$u_{n_k} \xrightarrow{k\to\infty} \bar{u} \in C^{0,\alpha}(D) \qquad \text{uniformly.}$$

As a result, we find that $\Omega = \{\bar{u} > 0\}$ is open, belongs to $\mathcal{A}$, and $\bar{u}$ solves (1) with $A = \Omega$. Moreover, it is clearly optimal because:

$$\mathcal{F}(\Omega) \leq \liminf_{n\to+\infty} \mathcal{F}(\Omega_n).$$

Idea of the Proof for $p \leq d$

(1) Minimizing sequences Ω_n tend to split further and further, ultimately converging in a suitable γ_p-sense to a relaxed solution. This is generally not a domain, but a p-capacitary measure. Namely,

$$\begin{cases} \mathrm{cap}_p(E) = 0 \implies \mu(E) = 0 \\ \mu(E) = \inf\{\mu(\Omega) \,:\, \Omega \supset E, \Omega \text{ } p\text{-quasi open}\} \end{cases}$$

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(2) If μ is a p-capacitary measure, we can define the Sobolev space $W^{1,p}_{\mu}$ of functions $u \in W^{1,p}(\mathbb{R}^d)$ such that:

$$\|u\|_{1,p,\mu} = \left(\int_D |\nabla u|^p \, dx + \int |u|^p \, d\mu \right)^{1/p} < \infty.$$

If f belongs to the dual, then the PDE (1) admits a unique solution $u_{\mu,f}$.

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If f belongs to the dual, then the PDE (1) admits a unique solution $u_{\mu,f}$.

(3) The mapping $(\mu, f) \mapsto u_{\mu,f}$ exhibits the following monotonicity properties:

$$\mu_1 \geq \mu_2 \implies u_{\mu_1,f} \leq u_{\mu_2,f} \quad \text{and} \quad f_1 \leq f_2 \implies u_{\mu,f_1} \leq u_{\mu,f_2}.$$

Idea of the Proof for $p \leq d$

(4) Existence of an optimal p-capacitary measure is established through lower semicontinuity and compactness. Then

$$\Omega = \{u_{\mu_{\text{opt}},f} > 0\}$$

is a p-quasi open set, $|\Omega| = m$, and it serves as a solution to the relaxation of the problem to p-quasi open sets contained in D.

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(5) When we further assume $f \in L^q(D)$, $q > d/p$, $m = |D|$, and a growth condition on the cost such as:

$$-g(x) \leq -C^{-1} f(x) \qquad \text{for every } x \in D,$$

then Ω above is open. The summability assumption on f cannot be removed, as counterexamples exist.

Finite Perimeter

Under mild assumptions, the perimeter of a solution Ω is finite, following the strategy from [Buc12] developed for the eigenvalue problem:

(1) Test the equation $\varphi = \bar{u} \wedge \varepsilon$ to obtain

$$\int_{\{0<\bar{u}<\varepsilon\}} |\nabla \bar{u}|^p \, dx \leq C_1 \varepsilon.$$

(2) Comparing $\bar{u}$ and the competitor $v := (\bar{u} - \varepsilon)^+$, we obtain:

$$|\{0 < \bar{u} < \varepsilon\}| \leq C_2 \varepsilon.$$

(3) By Hölder inequality, $\int_{\{0<\bar{u}<\varepsilon\}} |\nabla \bar{u}| \, dx \leq C\varepsilon$. Applying the coarea formula, we conclude:

$$\mathrm{Per}(\Omega) < \infty.$$

Related and Open Problems

- The full Dirichlet boundary condition in (1) can be replaced by the mixed Dirichlet-Neumann condition

$$u = 0 \quad \text{on} \quad D \setminus \Omega, \qquad \partial_\nu u = 0 \quad \text{on} \quad \partial D \cap \partial \Omega.$$

  This change allows for existence of open sets when $p > d$ and quasi-open sets when $p \leq d$, given that $m < |D|$ strictly.

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- The regularity of optimal shapes, except for the case $p = 2$, is currently an entirely open problem.

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Framework of the Problem

In [But+22], we focus on regularity of solutions to

$$\min \{\mathcal{F}(A) \ : \ A \subset D, \ A \text{ open}\}.$$

For simplicity, we consider the affine functional

$$\mathcal{F}(A) = -\int_D g(x) u_A \, dx + |A|,$$

where u_A is a solution to the PDE (here $p = 2$)

$$-\Delta u_A = f \text{ in } A, \qquad u_A = 0 \text{ on } \partial A.$$

Main Result

Theorem (Buttazzo, M., Mazzoleni, Tortone, Velichkov)

Let D be a bounded open set in $\mathbb{R}^d$, $d \geq 2$. Let $f, g : D \to \mathbb{R}$ be non-negative functions and assume that the following holds:

(a) $f, g \in L^\infty(D) \cap C^2(D)$;

(b) $0 \leq C_1 g(x) \leq f(x) \leq C_2 g(x)$ *for all* $x \in D$.

Then the free boundary of any optimal set $\Omega \subseteq D$ can be decomposed as

$$\partial\Omega \cap D = \mathrm{Reg}(\partial\Omega) \cup \mathrm{Sing}(\partial\Omega),$$

satisfying the following properties:

- $\mathrm{Reg}(\partial\Omega)$ *is locally* $C^{1,\alpha}$ *for some* $\alpha \in (0,1)$;
- $\mathrm{Sing}(\partial\Omega)$ *is closed and has Hausdorff dimension at most* $d-5$.

A few remarks on the assumptions

- The C^2 regularity assumption in (a) serves a technical purpose related to our use of second-order variations of J along vector fields.
- Assumption (b) plays a crucial role in establishing
  - **Lipschitz-continuity** of u_Ω, which follows from the outwards minimality of Ω and [Buc+15];
  - **non-degeneracy** of u_Ω, namely there is $\eta > 0$ such that

$$\|u_\Omega\|_{L^\infty(B_r)} \leq \eta r \implies u \equiv 0 \text{ in } B_{r/2}.$$

    which follows from the inwards minimality of Ω and [BV14].

  These properties are essential for subsequent blow-up analyses.

Is the Inclusion Condition Necessary for Regularity?

The inclusion constraint $\Omega \subset D$ is typically crucial for the existence of optimal shapes. However, in our context, it can be removed:

Theorem (Buttazzo, M., Mazzoleni, Tortone, Velichkov)

Let $f, g : \mathbb{R}^d \to \mathbb{R}$ be non-negative functions and assume that

$$f, g \in L^\infty \cap L^1(\mathbb{R}^d) \cap C^2(\mathbb{R}^d) \quad \text{and} \quad f(x), g(x) > 0 \text{ for all } x \in \mathbb{R}^d.$$

Then, there exists an open set $\Omega \subset \mathbb{R}^d$ that is a solution to

$$\min \left\{ -\int_A g(x) u_A \, dx + |A| \ : \ A \subset \mathbb{R}^d, A \text{ open and } |A| < \infty \right\}.$$

Morever, each such solution Ω is bounded so the regularity theorem applies.

Comparison between $f = cg$ and $f \neq cg$

- When $g = (2\lambda^2)^{-1} f$, our problem is equivalent to

$$\min_{u \in H_0^1(D)} \int_D \left(\frac{1}{2} |\nabla u|^2 - f(x)u + \lambda^2 \right) dx.$$

  For $f \geq 0$, $\partial\Omega$'s regularity follows from the one-phase Bernoulli problem's regularity theory ([AC81; CJK04; De 11; DJ09; JS15; Wei99]).

  If f changes sign, it becomes a two-phase Bernoulli problem, with recent regularity results in [DSV21; SV19].

- When $f \neq cg$, classical methods like truncations, harmonic replacements, and radial extensions in small balls are not applicable, making most free boundary regularity results impractical.

Adjoint State and Optimality Condition

(1) Introducing the adjoint function v_Ω as a weak solution of $-\Delta v_\Omega = g$ on Ω, with $v_\Omega \in H_0^1(\Omega)$, an integration by parts yields:

$$\mathcal{F}(\Omega) = \int_D \left(\nabla u_\Omega \cdot \nabla v_\Omega - g(x)u_\Omega - f(x)v_\Omega + \chi_\Omega(x)\right)\, dx.$$

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(2) For any smooth vector field $\eta \in C_c^\infty(D;\mathbb{R}^d)$ and $\Omega_t = (\mathrm{Id} + t\eta)(\Omega)$, if Ω is optimal, then the first variation is zero: $\delta\mathcal{F}(\Omega)[\eta] = 0$.

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(3) A computation shows that u_Ω and v_Ω are **formally** solutions to

$$\begin{cases} -\Delta u_\Omega = f \quad \text{and} \quad -\Delta v_\Omega = g & \text{in } \Omega, \\ u_\Omega = v_\Omega = 0 \quad \text{and} \quad |\nabla u_\Omega||\nabla v_\Omega| = 1 & \text{on } D \cap \partial\Omega. \end{cases} \tag{4}$$

Epsilon-Regularity Theorem for Viscosity Solutions

We say that $(\Omega, u_\Omega, v_\Omega)$ is a viscosity solution to (4) if for all $x_0 \in \partial\Omega \cap D$ at which Ω admits a one-sided tangent ball there are

- a unit vector $\nu \in \mathbb{R}^d$;
- constants $\alpha, \beta > 0$ such that $\alpha \cdot \beta = 1$;
- a decreasing sequence of radii $r_k \to 0$;

such that the blow-up sequences u_{x_0,r_k} and v_{x_0,r_k} converge uniformly in every ball to the blow-up limits

$$u_0(x) := \alpha(x \cdot \nu)_+ \qquad \text{and} \qquad v_0(x) := \beta(x \cdot \nu)_+.$$

Epsilon-Regularity Theorem for Viscosity Solutions

The viscosity formulation happens to be the correct framwork. Indeed, in [MTV23] we prove the following epsilon-regularity theorem:

Theorem (M., Tortone, Velichkov)

Let $u_\Omega, v_\Omega : D \to \mathbb{R}$ be continuous and non-negative functions and assume that

- $\Omega = \{u_\Omega > 0\} = \{v_\Omega > 0\}$;
- $(\Omega, u_\Omega, v_\Omega)$ *is a viscosity solutions to* (4);
- u_Ω *and* v_Ω *are* ε*-flat in a ball* $B_r(x_0) \subset D$ *centered on* $\partial\Omega$.

Then the free boundary $\partial\Omega$ *is* $C^{1,\alpha}$*-regular in* $B_{r/2}(x_0)$.

Outline of the Proof: $C^{1,\alpha}$-Regularity

Let $\Omega_t := (\mathrm{Id} + t\eta)(\Omega)$. If Ω is optimal, then it is a **stable critical point** of the shape functional $A \mapsto \mathcal{F}(A)$:

$$\frac{\partial}{\partial t}\Big|_{t=0} \mathcal{F}(\Omega_t) = 0 \qquad \text{and} \qquad \frac{\partial^2}{\partial t^2}\Big|_{t=0} \mathcal{F}(\Omega_t) \geq 0.$$

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This notion is stable under blow-up limits at $x_0 \in \partial\Omega \cap D$:

$$u_0(x) := \lim_{n\to+\infty} \frac{u_\Omega(x_0 + r_n x)}{r_n} \quad \text{and} \quad v_0(x) := \lim_{n\to+\infty} \frac{v_\Omega(x_0 + r_n x)}{r_n}.$$

So $\Omega_0 = \{u_0 > 0\}$ is also a stable critical point with $f \equiv g \equiv 0$.

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So $\Omega_0 = \{u_0 > 0\}$ is also a stable critical point with $f \equiv g \equiv 0$. The Boundary Harnack Principle [MTV22] allows us to conclude that:

$$u_0/v_0 : \Omega_0 \to \mathbb{R}$$

is Hölder-continuous in Ω_0 up to the boundary.

Outline of the Proof: $C^{1,\alpha}$-Regularity

A further blow-up in zero yields Ω_{00} stable critical point and $u_{00} = \Lambda v_{00}$. Thus, u_{00} is a stable critical point for the Alt-Caffarelli's one-phase functional

$$\mathcal{G}(u, B_R) = \int_{B_R} |\nabla u|^2 \, dx + |B_R \cap \{u > 0\}| \qquad \text{in every } B_R \subset \mathbb{R}^d.$$

Outline of the Proof: $C^{1,\alpha}$-Regularity

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$$\mathcal{G}(u, B_R) = \int_{B_R} |\nabla u|^2 \, dx + |B_R \cap \{u > 0\}| \qquad \text{in every } B_R \subset \mathbb{R}^d.$$

A monotonicity formula is available for the one-phase problem, so a third blow-up

$$u_{000}(x) = \lim_{k \to +\infty} \frac{u_{00}(r_k x)}{r_k}$$

is 1-homogeneous stable critical point for the Alt-Caffarelli's functional $\mathcal{G}$.

Outline of the Proof: Hausdorff dimension bounds

The proof of $\mathcal{H}^{d-d^*+\varepsilon}(\mathrm{Sing}(\partial\Omega)) = 0$ proceeds as follows:

(i) The bounds on d^* for minimizers of $\mathcal{G}$ rely on the well-known Caffarelli-Jerison-Kenig stability inequality [CJK04].

However, this inequality is challenging to apply when passing to blow-up limits and developing a dimension reduction principle.

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(ii) Our notion of **stable critical point** allows to develop a dimension reduction principle and that there is a critical dimension d^*.

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(ii) Our notion of **stable critical point** allows to develop a dimension reduction principle and that there is a critical dimension d^*.

(iii) Our stability notion is equivalent to the stability inequality on smooth cones (cones with isolated singularities), leading to the bounds $d^* \geq 5$ [DJ09] and $d^* \leq 7$ [JS15].

Thanks for your attention!

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